

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI
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BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI
(RAJASTHAN)
WILP Division

Organization: Walmart Global Tech India Pvt. Ltd. **Location:** Bengaluru, Karnataka, India

Duration: Approximately four (4) months

Date of Start: 01 April 2026

Date of Submission: 02 August 2026

Title of the Project: Real-Time Social Media Mining and Trust-Aware Sentiment Analysis using Large Language Models for Retail Product Feedback Optimization.

ID No. / Name of the Student: 2020AA05641 — Vishal Singh

Name(s) and Designation(s) of your Supervisor and Additional Examiner:

- **Supervisor:** Mr. Varunendra Pratap Singh, Principal Software Engineer, Walmart Global Tech, Bengaluru.
- **Additional Examiner:** Mr. Suhas Joshi, Principal Engineer, Walmart Global Tech, Bengaluru.

Faculty Mentor: Ms. Pradnya Kashikar, BITS Pilani — WILP Division.

Key Words: Sentiment analysis; ModernBERT; aspect-based opinion mining; zero-shot NLI; mul-timodal vision-language models; trust scoring; Reddit; retail feedback.

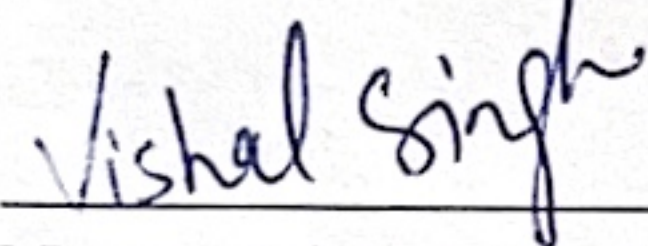
Project Areas: Applied Natural Language Processing; Multimodal Machine Learning; Data Engi-neering; Human-in-the-Loop Analytics.

Abstract:

Every day, customers, store associates, and Walmart delivery drivers describe what actually happens inside retail — not on the retailer’s app, but on Social Media(Reddit) — yet almost none of that signal is captured today: off-the-shelf sentiment services choke on sarcasm, treat every post as equally trustworthy, and never touch the screenshots that carry half of the evidence. This dissertation closes that gap with *Retail Senti-ment Intelligence* (RSI), an offline-first, zero-inference-cost pipeline that turns the raw Reddit stream into a trust-weighted, aspect-tagged, priority-ranked feed an analyst can act on in minutes. Posts flow in from **32 retail-relevant subreddits** and receive a 0–1 trust score whose three components (account metadata, an LLM-graded credibility signal, and a retail-insider lexicon) are exposed to the analyst so nothing is silently dropped. A **ModernBERT** sentiment head is fine-tuned through a three-stage curriculum ending in **200 hand-labelled posts**; a zero-shot **DeBERTa-v3 NLI** classifier tags each post against a fixed eight-aspect retail taxonomy; and **Gemma3 4B** handles screenshots through a multi-pass anti-hallucination prompt (describe → OCR → discard

captions whose entities are absent from the OCR). A React/FastAPI dashboard exposes Brand Health, Alert Feed, Post Explorer, a Kanban Post Lifecycle, Insights and Competitor Analysis, and Notifications, backed by a human-in-the-loop *Review & Validate* workflow, a dual-draft (rule + LLM) smart-reply composer, and a learning-loop store that becomes the next round of training data. Evaluated 5-fold out-of-fold on the 200-post gold set, sentiment macro-F1 climbs from a Twitter-RoBERTa baseline of **0.6272** to **0.7642** overall, with the largest gains on long posts the baseline could not model. On a 32-image benchmark the multi-pass vision pipeline drops hallucination from **50% to 0%**, lifts text extraction from **25% to 75%**, and retail-signal recall from **40% to 81%**; the trust module correctly flags **12 of 15** low-trust posts confirmed by human annotators. All inference runs locally, so the reference pipeline carries **zero commercial API cost**, uses only public data, and ships as a reproducible artefact backed by SQLite (WAL) and a JSONL cost log.

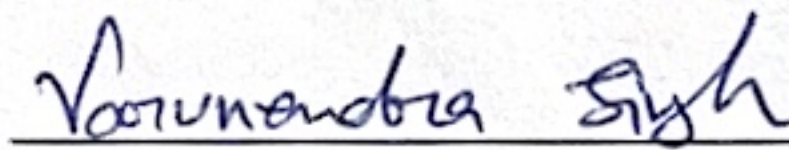
Signature of Student



Name: Vishal Singh

Date: 01-08-2026

Signature of the Supervisor



Name: Varunendra Pratap Singh

Date: 01-08-26